**Power connector (2 x 10-pin)** *Usage: Optional*

| | A | Dir | | | B | Dir | |
|---|---|---|---|---|---|---|---|
| 1 | IN+ | I | Power positive input | 2 | IN+ | I | Power positive input |
| 3 | IN+ | I | Power positive input | 4 | IN+ | I | Power positive input |
| 5 | IN+ | I | Power positive input | 6 | OUT+ | O | Power positive output |
| 7 | OUT+ | O | Power positive output | 8 | OUT+ | O | Power positive output |
| 9 | OUT+ | O | Power positive output | 10 | OUT+ | O | Power positive output |
| 11 | OUT- | O | Power negative output | 12 | OUT- | O | Power negative output |
| 13 | OUT- | O | Power negative output | 14 | OUT- | O | Power negative output |
| 15 | IN- | I | Power negative input | 16 | IN- | I | Power negative input |
| 17 | IN- | I | Power negative input | 18 | IN- | I | Power negative input |
| 19 | SENSE- | I | Sense negative input | 20 | SENSE+ | I | Sense negative input |

Used on slots #1, #2 and #4 only

Examples of backplane connector models: PH2-20-UA, Examples of backplane connector models: PH2-20-UA, 77313-124-22LF, ZL202-20G, 10129381-922002BLF, ZL202-20G

**Signal connector** *Usage: Mandatory*

| | A | Dir | Description | | B | Dir | Description |
|---|---|---|---|---|---|---|---|
| 1 | PE | I | Protective Earth | 1 | +12V | I | DC power |
| 2 | +12V | I | DC power | 2 | +12V | I | DC power |
| 3 | +12V | I | DC power | 3 | GND | | |
| 4 | GND | | | 4 | +3V3 | I | DC power |
| 5 | +5V | I | DC power | 5 | +3V3 | I | DC power |
| 6 | +5V | I | DC power | 6 | +VAUX | I | Backup DC power |
| 7 | GND | | | 7 | GND | | |
| 8 | SHUTDOWN | I | Shutdown/Low-power (active low) | 8 | RESETn | I | Module reset (active low) |
| 9 | SMBUS_ALERT | O | Shared SMbus alert (active low) | 9 | NFAULT | I/O OD | Fault (open drain) |
| 10 | SCL/SMBCLK | I | Shared I²C/SMbus Clock | 10 | SDA/SMBDAT | I/O | Shared I²C/SMbus Data |
| 11 | GND | | | 11 | GND | | |
| 12 | GND | | | 12 | GND | | |
| 13 | RSVD1 | | Reserved / Ethernet TXRX_A- | 13 | OE_SYNC/TINn | I | Module Trigger In |
| 14 | RSVD2 | | Reserved / Ethernet TXRX_A+ | 14 | AC_FREQ | I | AC Mains frequency |
| 15 | RSVD3 | | Reserved / Ethernet TXRX_B- | 15 | TOUTn | O | Module Trigger Out |
| 16 | RSVD4 | | Reserved / Ethernet TXRX_B+ | 16 | REFCLK+ | I | SE or CLK+ 10MHz reference |
| 17 | RSVD5 | | Reserved / Ethernet TXRX_C- | 17 | REFCLK- | I | CLK- 10 MHz reference |
| 18 | RSVD6 | | Reserved / Ethernet TXRX_C+ | 18 | UART_RX* | O | Shared UART RX |
| 19 | RSVD7 | | Reserved / Ethernet TXRX_D- | 19 | UART_TX* | I | Shared UART TX |
| 20 | RSVD8 | | Reserved / Ethernet TXRX_D+ | 20 | BOOTn** | I | Module bootloader select |
| 21 | SCLK | I | Shared SPI CLK | 21 | GND | | |
| 22 | MOSI | I | Shared MOSI | 22 | PCIe_CLK+ | I | PCIe clock |
| 23 | MISO | O | Shared MISO | 23 | PCIe_CLK- | I | |
| 24 | IRQn | O | Module IRQ | 24 | GND | | |
| 25 | CSAn | I | Module Chip select A | 25 | PCIe_RX- | I | PCIe RX |
| 26 | CSBn | I | Module Chip select B | 26 | PCIe_RX+ | I | |
| 27 | HSD+ | I/O | USB 2.0 | 27 | PCIe_TX- | O | PCIe TX |
| 28 | HSD- | I/O | | 28 | PCIe_TX+ | O | |
| 29 | SSTX+ | O | USB 3.0 Tx | 29 | PCIe_CLKREQ | I | PCIe clock request |
| 30 | SSTX- | O | | 30 | PCIe_WAKE | I | PCIe wake |
| 31 | SSRX+ | I | USB 3.0 Rx | 31 | GND | | |
| 32 | SSRX- | I | | 32 | ETH_SYNC | O | IEEE 1588 sync |

*) Connect module UART_RX to master MCU UART_TX and module UART_TX to master MCU UART_RX

Examples of backplane connector models (gold plating of 0.25 um or thicker, not „flash”): 10018783-10101TLF, 10018783-11101TLF

**) Requires a “pull-down” resistor (e.g., 4.7 kΩ to 10 kΩ) used to detect the presence of the module